

Assignment #1 (10%)

SBEG3603 Visualization & System Customization for Geoinformation An Application Programming Interface (API) Development

Submission date: Once completed

Objectives of the assignment:

- To expose students to practical spatial data preparation and database design workflow.
- To develop understanding of how raw tabular and spatial datasets are cleaned and prepared for system integration.
- To build a secure API backend using PHP for geospatial data access.
- To strengthen awareness of data type correction, validation, and error handling in application development.
- To train students to use AI-assisted prompting for code generation, code conversion, refinement, and debugging.

Data:

You are given a set of spatial and non-spatial datasets related to parcel, locality, and city-related records. These datasets include:

- DBKU_Locality.shp
- KCH_DCDB_Parcel_Polygon_70.shp
- MK7_MK8 - city.csv

The datasets may contain issues such as:

- incorrect data types,
- text-based currency values,
- number fields stored as strings,
- inconsistent dates,
- missing values,
- non-standard naming conventions.

Before a web GIS application can be developed, the data must first be prepared properly in a spatial database and exposed through a secure API backend.

Task:

You are required to develop an API as the backend foundation for the later web GIS application in Assignment #2.

1. **Prepare and integrate datasets** (CSV and shapefiles) into a PostgreSQL/PostGIS database, including correcting data types (e.g., currency, numeric, date).
2. **Design API endpoints and define data structures** aligned with the dataset and expected application usage.
3. **Implement full CRUD functionality:**
 - **Create:** Insert new records into the database

- **Read:** Retrieve and display records (read all records, read single record, or filter one or more attributes (e.g. locality code, zone code, parcel identifier, other suitable fields))
 - Spatial-ready response structure by returning such as coordinates, geometry representation, or a GeoJSON-like structure.
 - **Update:** Modify existing records
 - **Delete:** Remove records safely with proper checks
4. **Develop PHP scripts** to handle API requests, including data validation, processing, and structured JSON responses.
 5. **Apply security and error handling mechanisms**, including input validation, prepared statements via PDO, and clear exception responses.
 6. **Utilize AI-assisted prompting** to support API development and document the prompts used (vibe-based or refinement-based).
 7. **Test and validate API functionalities** to ensure accuracy, consistency, and protection against common security risks.

Note:

- Add your working folder containing all PHP files into a zip (zip), or RAR (rar), or 7z (*.7z) file. Rename the compressed file to **yourname_assignmentnumber** (i.e., **ivinamri_assignment01.zip**).
- Upload the zipped file to the eLearning upload tool.
- Failure to submit this assignment will be automatically subject to a mark of zero (0).